No attempt is made to teach these children anything. There are no schools or teachers. Their teacher is the multi-dimensional environment that is designed to interest, stimulate, and challenge. The basic information they need has been implanted in their supplementary brains. It has been found that any attempt to teach them only retards the learning process. Experiments have shown that the best way is to let these children explore their environment. If you want a child to learn to work a device, you put it near him. He does the rest. His natural curiosity leads him to observe the operation of the teleprojection screen, and he begins to request Corcen to furnish him with programs. These cover the entire range of knowledge and entertainment. However, obscene material used on TV and movies in the previous century, showing brutality, murder, and sadism, are not available in the nursery.

When a child observes things that correlate with information programmed into his supplementary brain, a flash of insight comes. He is literally on fire intellectually. His thoughts race into the exciting new areas of thinking and feeling that he is discovering inside himself. He learns that the spirit of creative inquiry is one of the most delightful things that he can experience. A great feeling of dignity and worth is achieved as the child explores his own inner resources and integrates them into his expanding world of people and things!

Children are not informed about the areas in which their brains have been pre-programmed. They discover these for themselves. The greatest thrill comes when they discover the one area in which they have total information. No other human knows the pre-programming of their supplementary brains. These patterns have been set up by Corcen to add spice and adventure to life. "Scientific research has established that the curiosity of a human child is many times the amount needed for his intellectual development if environmental conditions are stimulating and there are no teachers to interfere," Scott's associate tells him. "Everyone who associates with growing children is instructed to avoid telling them what they should or shouldn't do. Back in the twentieth century, education was sometimes considered a process of helping a child fit into society. Now we know that fitting into society may be taken for granted, for we have found that children raised without hostility and scarcity develop social skills that enable them to achieve the finest possible relations with other people.

"Those who associate with children think only of understanding the feelings and interests of the child. They ask the children questions and practically never give them answers. The children have to find their own answers—perhaps from the limitless facilities of Corcen. This makes life more exciting and never blighting.

They develop a feeling of intellectual adventure. The child develops with personal authenticity."

The Creative Adventure of Educational Research

The men and women who enjoy the challenge of improving the next generation have developed thousands of new ways to meet the needs of infants and children. Nothing is taken for granted. Little is copied from the past. The people who are working in this area of the new civilization do not feel they have any final answers. They know that they are getting results. They are confident that their methods are superior to any ways of rearing children that have ever been used before in the history of the world. They

know that by observing carefully and thinking creatively and by continually measuring the results, they will find more effective ways of doing things. Whatever "errors" they are now making will eventually be corrected.

Through sensitive, scientific research an effective environment for each age level has been evolved. Constant research is in progress to find out just what is happening and how it can be improved. It has been found that an environment that gives optimal enjoyment and development at six months retards at one year. The surroundings that are best for a one year old will hold back a two year old, and so forth. Great care is taken to provide an environment that is scaled to meet the needs of each individual age level.

"The nurseries are designed so that the child never needs correction, for it can do nothing undesirable in this environment," Scott is told. "In the twentieth century a toddler of two years old could hardly do anything right. Every time it turned around it had to be admonished, 'No! Don't go out in the street, you'll be killed. Don't reach up to the top of the dresser because you might upset Mommy's bottle of perfume. Don't pull the tail of the dog, or he might bite you,' and so forth. Such constant bombardment of a young child makes him a lifelong slave to external patterns."

The Cybernated Nurseries

Scott sees that the nurseries of young children in the twenty-first century are scaled down in size so that children have no feeling of being small and inferior. All natural functions, such as eating, eliminating, playing, sleeping, and so forth, can be done in any way that the child chooses in the cybernated nursery. Living areas are engineered in such a way that children can not hurt each other prior to the time that they develop their feeling of empathy for all living things. These babies grow up in an atmosphere free from hostility, criticism, deprivation, scarcity, and jealousy. They are thus able to develop positive feelings of co-operation and comradeship for all other human beings that were impossible for individuals who lived in previous centuries.

Food is eaten when desired. There is no pattern of breakfast, lunch, and dinner, such as was common in previous centuries. This three-times-a-day eating habit was probably designed for the convenience of cooks more than the needs of individuals. Scientific research has shown that the human body operates best when it snacks on nutritious foods at frequent intervals. The eating of three large meals results in biochemical reactions that are not consistent with the highest level of health.

Scott knows that in previous centuries mealtime was often a struggle between mother and child. "Just taste it—you may like it. Don't hold your fork that way—it's not polite. You haven't finished your plate. You've dropped food all over your shirt." It was often a nuisance for a mother to feed a young child in the past, and she sometimes became impatient doing it. Even an infant senses feelings of impatience and hostility, and thus the seeds of insecurity and fear were planted in his personality.

Up until the twenty-first century most of the ways of handling children were based on the needs of parents and adults. For example, when men and women experimented to find a better way to feed a twenty-month-old child, they found that the child enjoyed putting bite-sized nuggets in its mouth and

sucking food from the nipple-like spouts. The children at this age level also enjoyed pressing a pad which would release small wafers with an accompanying musical tone. Feeding systems were set up that enabled the children to eat whenever they wished. The children usually were noisily exuberant, but there were no adults to be annoyed. And there was no mess to be cleaned up by weary, harried mothers—the cybernated cleaning mechanisms were on the job. Eating was always fun!

In the past bathing young children was sometimes a nuisance and irritation to both mother and child. Men and women of the twenty-first century asked themselves, "How do you set up a bathing situation so that young children are automatically attracted to it?" They didn't want adults to have to brow-beat the youngsters to tell them it was time to get their bath. They wanted the bathing situation to fit in with the needs and interests of the child as felt by the child. They wanted the child to clean himself and get a bath simply because he wanted to. But how do you make an eighteen-month-old child want to take a bath? They found that they had to make bathing pleasant. After many experiments they discovered that a six-inch pool of warm swirling water with random shower sprays that give a nice feeling on the skin is most effective.

Scott laughs as he watches the children enjoying the cybernated bath. A screen in the bottom of the pool automatically comes up if a child's head goes under water. When the child gets tired of playing in the water, he can either dry naturally in the warm air or lie down on a rocking towel couch that rolls him over and over. Sometimes children lie in these rocking towel couches just because they enjoy them—whether they need drying or not.

Toilet training is also simplified to make it pleasant for the child and free of menial activity for adults. A young toddler can urinate and defecate at any time or place in his specially designed environment. The cybernators watching over the children immediately sense wetness and an automatic, roving, cleaning mechanism cleans up the floor and child. Since no fuss or guilt arises over toilet training, the children learn to use the toilet mechanisms at an earlier age than in the past.

"An amazing discovery has been made in these cybernated nurseries," Scott's associate tells him with great enthusiasm. "The inferiority complex, which psychologists and psychiatrists had regarded as a basic part of human personality, does not develop! We don't destroy their feeling of worth during their defenseless childhood. This may be the first era in human history that has produced confident, secure humans with no impediments to achieving the greatest joy of living."

Learning By Self-Directed Living

Scott observes the series of environments that enable the infants and children to develop to the fullest in every way. It has been found that a graded series of twelve environments is needed to develop the newborn to the age of five. After the two-year level is reached, the child decides for himself when to go to the next environment. It is not considered "bright" for a child to push himself to an advanced environment as long as he is comfortable and interested in the current environment.

The advanced environments for the older children have equipment and facilities that would have been

unavailable even at university levels in the previous century. Teaching machines have been designed to attract and hold the interest of the children. Three-dimensional teleprojections of every kind are available through Corcen at any time of night or day. All activity is self-motivated. There are no classes, no teachers, and no tests. The educational researchers are constantly amazed at the intelligent self-direction of these young children. They learn more rapidly when left alone in their specially designed environments than any previous children who were put in large boxes called schoolrooms, complete with Miss Brooks to spoon feed information and then force them to regurgitate it at examination time.

These children are never subjected to criticism, for it has been found that criticism represses and reduces their potential. They are surrounded by constructive examples in place of criticism. Their egos do not need the amount of praise that was so effective in teaching in the past. Each is free to meet life on his own terms and to learn to express his emerging uniqueness. Perhaps no previous society could let children develop so fully as individuals and at the same time provide them with a cultural heritage of such enormous richness.

There are few adults in the cybernated environments of the children. The adults who are there have chosen to spend time with children for one reason only—they enjoy it. They never act as disciplinarians since the cybernated environments are so designed that no child can hurt himself or others. A relaxed companionship of a quality that never existed before between parents and children develops between these adults and the children.

The Greatest Research Program

"Previous centuries were eras of scarcity," Scott's associate reasons, "and this scarcity was not only in material goods. Few children in the past felt enough love, warmth, security, feeling of worth, and freedom to develop in their own way. They were stifled by criticism, comparison, and censure. Only in the twenty-first century has the creative intelligence of man solved these problems. At last, little children appear to have their needs met. But each decade shows that further improvement is possible.

"One of the most continuous needs of a child is that of a feeling of security, of human closeness, and of rapport with a friendly world." Scott watches a group of young infants in their cybernated cribs as his companion leads him through the nursery. "The most intensive research program in human history was launched to give effective solutions to the enormous problems in designing cybernated environments for young children that were superior in all respects to the traditional family and home. How can a soothing cybernated voice establish rapport with an infant? How can a three-dimensional, teletactile arm that is felt by an infant in his crib be activated to give even more security and feeling of constancy than a mother's arm of the previous century? How can cybernated mechanisms be designed to give children more warmth and more of everything they need than even a superior mother could give in previous times? How can machines form the link between words and things so that language habits are created? What types of situations do these twenty-first-century infants and children need to develop a tolerance for frustration—to develop patience and calmness when things do not go as expected? To what extent can the older children be relied on to act as models for younger ones? To what extent are adult models needed in the environment of infants and children at various ages? What is the best way to give a child experience with the adult environment so that he can acquire an independence and feel at home in the world? Bit by bit, research yields answers to these and countless other problems.

"No twentieth-century answers to these questions are applicable in our world. Overall conditions have changed too much to use the old 'wisdom.' For the first time in the history of man, creative, scientific intelligence is applied to the problems of giving children the maximum of everything they need to develop satisfying, purposeful lives. It took decades to arrive at the preliminary patterns of the cybernated nursery. But perhaps even greater change awaits us in the future, for the nurturing of the young is the foundation of every civilization."

An Age of Individuality

On the wall in the conference room, a large portrait appears. Below the frame Scott reads the words of the scientist that founded this laboratory in 2014:

> *Our society is patterned for individuals. All social structures and physical arrangements are designed to meet the needs of individuals and accommodate almost any diversity. We do not feel that children should do anything other than what they select themselves. "Life, Liberty, and the Pursuit of Happiness" applies equally to children.*

The tour of the cybernated nurseries is almost over. Scott is impressed by the changes that have occurred in the nurseries since he was a child. "Learning is a lifetime process that starts at the time the supplementary brain is implanted in the growing nervous system of the embryo," his co-worker points out. "Education is only stopped by death. There is no graduation or diploma that artificially chops up the learning process.

"The only thing comparable to graduation usually occurs when a child is about five years old. When the child's interaction with Corcen shows that he can safely be permitted to leave the nursery area, he is welcomed as a full member of society. He is then entitled to his own apartment. He begins to make his own choices of what he wants to do and where he wants to live."

"Youngsters are amazing," agrees Scott. "There are eight-year-old girls that travel to the moon and live there several years. I knew a seven-year-old boy who was invited to join the crew of a space ship."

"Children, however," Scott's associate continues, "are not motivated to develop into adult roles at any particular age. When it happens, it happens. No one is watching them. No one measures them. No one compares them. No one is worried if they lag. No one pushes them to "get ahead." Each individual feels completely free of all pressures to do anything." He flashes a knowing smile, "And, you see, this makes them want to do everything!"

11.A Visit to Corcen

While Scott is in India, Hella remains at the underwater apartment in the Exumas. There is so much to explore, both inside her brain and outside. Her supplementary brain has been programmed by Corcen with a Ph.D. level of information on oceanography and marine science. Her life has been so busy in other areas that she has never used this information except in a peripheral way. She is swept up in the

fascinating correlation between the facts and theories that were quietly stored in her brain and the marine world around her. She asks Corcen to send her data to bring her up to date. By using her own inner resources and through discussions with others who have backgrounds in depth in this area, Hella spends the better part of a year in one of the keenest of human pleasures—the intellectual experience of integrating the world outside the brain with the information and knowledge inside. By color teleprojection, Scott and Hella mutually share their experiences and feelings—often hourly when something exciting is happening.

It has been many decades since Hella has visited Corcen and the North American Cybernated Industrial Complex. When she was five years old, she visited these centers that play a primary role in providing a good life. She knows that returning now would give her a heightened perception and a depth of meaning that was not available to her as a young child. For a number of years, she has been thinking how worthwhile it would be to visit Corcen, but other activities have always clamored louder for her attention. Now a group of men and women who have been enjoying the city in the sea is headed in this direction, and Hella decides to go with them.

One of the nicest things about living in the twenty-first century is the enormous range of choices. It is far greater than any other civilization was able to offer its citizens. Although Hella and her friends are in a relatively isolated area, they have the choice of a sailboat such as Scott used or an automatically powered seacraft. They could call for a variety of aircraft, depending upon their needs and how far they want to go. They could board a submarine freighter that services island communities, or they could use the GEM (Ground Effect Machine).

Since they want to spend a day roaming through the islands of the Bahamas, they choose the GEM. It can travel above a more or less flat surface at speeds up to 200 miles-per-hour. This machine skims at a height of about four feet above any surface. Whether water or land, a paved road or a rough field—it doesn't matter. It is supported in the air by three circular jets of air directed downward toward the ground. With relatively little energy this "ground effect" is able to lift a heavy craft in the air just far enough for the air to escape around the edges of the three ring vortices. As long ago as 1950, the British developed a GEM that skimmed the English Channel between Britain and France.

For a day Hella and her friends use the GEM to cruise around the Bahama Islands. The clear water and colorful coral banks are still engaging. They skim by Eleuthera, Abaco, Nassau, An-dros, the Berry Islands, and, finally, Bimini at sunset. Then off they go across the Gulf Stream to Miami.

Hella and her friends leave the GEM at the South Florida terminal the next morning and board the linear-acceleration train that will take them to Corcen in the Rocky Mountain area of the United States. Within the gleaming metal walls of these enormous trains, attractive living areas permit a continuation of the living patterns of one's home. This train travels in a large tube with a partial vacuum. It has no engine and no wheels. It is electrodynamically supported above a V-shaped rail that electromagnetically propels each car of the train. The negative charge on the probe projecting from the front repels moisture and dust particles in the tube ahead. This diminishes resistance and allows the train to develop a 2,000 mile-per-hour speed. Yet, even when accelerating or stopping, it feels as steady as a concrete building.

As Hella's train speeds through Florida, it passes through a cybernated farm over 200 miles long and 50

miles wide. Tracks 100 feet wide run the entire length of the farm. Large cybernated mechanisms slowly travel up and down these tracks to prepare the land for planting, to place seeds, fertilize, and water them. On a return trip several days later, these plants will be watered and cultivated, if needed. At the right time the vegetables will be picked, quick-frozen, and packaged by the enormous farming machines. Weather control eliminates all losses from freezing, drought, or floods. The cybernated Florida farm complex supplying food for one-fourth of a continent does not need a single human in attendance. Corcen co-ordinates the actual operation through a local cybernator programmed for scientific farming. The one-hour journey passes rapidly as Hella and her friends enjoy a wide range of cybernated entertainment.

The World Correlation Center

Corcen is housed 2,000 feet below the top of a large mountain to give it protection from any meteorite that might survive the fall into the earth's atmosphere. Many thousands of years ago a large meteorite plowed into Arizona and left a crater 3,870 feet wide by 560 feet deep. Radar stations throughout the globe constantly scan space for large meteors. On the rare occasion when a dangerous meteor is detected, missiles are sent out to pulverize it while it is still thousands of miles from earth.

Hella, standing at the entrance to the mountain, surveys the countryside below. She thinks briefly of today's missile technology and is glad it is used to protect mankind from remote dangers instead of threatening and killing.

A combination of high-speed elevators and escalator-type sidewalks convey Hella and her friends to the underground Corcen complex. Hella enters a large room where a six-foot sphere is electrodynamically suspended ten feet above the floor. Hella looks at this sphere with a feeling of awe and appreciation.

This is Corcen—the master computer that correlates the interactions of all people and all automated machines throughout the entire world.

In the time it takes her to draw one breath, this remarkable servant of man has probably made ten billion decisions based on the scanning of trillions of bits of information. If everyone on earth were superbly organized into a tremendous bureaucratic complex, it would be impossible to do in a year what this computer can perform in a second. Corcen does a zillion times more for each individual than any government of the past could possibly have done.

A teleprojection of a human guide now speaks to the visitors to Corcen. "This computer that we call 'Corcen' is our servant, not our master although it is a servant whose abilities far exceed ours. Its only purpose is to free us of routine problems and permit us to live in our own way. It responds quickly to criticisms and suggestions from all individuals, checking them out and seeing what can be done about them. No elected politician of previous times responded as consistently and effectively to the needs of his voters. Corcen never tells us how we should run our lives. It simply specifies that if we want certain results, we should go about getting them in certain ways."

"If anyone doubts who is in supreme command in this man-machine complex, let him consider this: the master power switch that could inactivate Corcen is on the wall over there. We would also have to shut

down the emergency duplicate of Corcen that is maintained in Europe. If these switches were thrown without planning, it would result in utter chaos. With careful planning it would be possible to inactivate Corcen and fragment the world once again. We could chop up the world into as many pieces as we wanted and operate each one independently. Confusion and a return to the primitive society of our ancestors would be the unhappy result. If we asked Corcen to plan its own elimination for us, it would probably be done in a way that would minimize the disadvantages. But it would be like killing what makes us truly human, truly free, truly happy."

"How completely we trust ourselves and each other," thinks Hella, looking at the master switch that activates Corcen. "No one will ever touch it, but it's nice to know it's there."

Ultimate Predictability

The teleprojected figure walks to a large table. The group follows him. A loudspeaker above the table begins, "The fundamental principle upon which Corcen operates is that decisions with a high degree of predictability can be made when adequate facts are available. We would like to give you a demonstration of this. Before you is a table twenty-four feet in diameter. Above the table you will see a small container that has fifty steel balls that are exactly twelve millimeters in diameter. These balls will now be mixed.

The transparent container holding the fifty balls turns upside down so that all balls jostled around to a new position.

"You will notice that there are electronic sensors in twelve positions surrounding these balls," the loudspeaker continues. "These sensors in a millionth of a second have already determined the location of each of these balls. This data will now be fed into Corcen (here Hella looks up at the six-foot sphere, only fifty feet away, and less than a second later Corcen will predict the eventual landing place of each of these balls when dropped onto the table three feet below."

The group looks at the table and notices that there are a number of white dots that have appeared on the dark surface. The speaker continues to describe what everyone has now guessed. "When these balls are dropped, they will tumble against each other; they will hit the table; they will bounce. Some of them will hit other balls while rolling, but within a few seconds all of these balls will stop rolling. They will come to rest exactly on the fifty white dots. Let us see if Corcen has accurately predicted their behavior."

The transparent container releases its flood of fifty balls. There is a series of metallic clanks as these fifty steel spheres bounce around in apparently random fashion. But within twelve seconds all but one come to rest. One seemingly erratic ball has bounced against the edge of the table. It hits a stopped ball that rolls to cover a white dot and, in so doing, deflects its own movement until, at last, the only dot remaining is covered. Hella takes a deep breath. Fantastic predictability!

The speaker continues. "If we provided our best mathematician with pencil and paper, it would have taken him over ten years to make the computations with the same degree of accuracy that Corcen produced in less than a second!

"This is the guiding principle of our Correlation Center: although we know it is impossible to predict the behavior of *single* atoms, the prediction of the average behavior of an *aggregate* of atoms—which we

regard as objects in the real world—is predictable within stated limits of reliability if we have an adequate sampling of facts. Note that we do not have to know all of the facts—this is impossible. An adequate sampling of relevant facts does the job for us."

The speaker above the table stops. The teleprojected guide again takes over. "Several centuries ago the affairs of mankind were far simpler." A large screen lights up. It shows a map of the world with population figures superimposed on each country. "Up until the eighteenth century most European nations had less than 25 million people. Great Britain never exceeded ten million people. The interaction of government with economic and social affairs was relatively simple. Most intelligent citizens who wished to be informed of the issues could have made fairly useful predictions. If predictions were inaccurate, the stakes were not high, and things tended to move slowly. They could have been corrected by the succeeding generation without too much harm to mankind.

"The situation changed radically in the twentieth century. Prior to that time a war might have killed a half-million people. The first major war of the twentieth century killed ten million people. The second major war killed five times as many. Had a third major war taken place, it is likely that billions of people would have been killed, to say nothing of the devastated cities and industrial plants. An ulcer in the stomach of a dictator could have played a part in a hasty emotional reaction that could have caused this catastrophe. Capricious, ineffective government by individuals could no longer have been tolerated. During most of the twentieth century, a mixture of democratic and totalitarian governments controlled the lives and destinies of over 100 petty, touchy, nationalistic units. No one felt secure."

Hella closes her eyes momentarily. She is glad she didn't live in such troublesome times.

"About the middle of the twentieth century, the electronic computer was born," the teleprojected guide continues. The large screen continually changes to illustrate the thought of the guide. "At first it was a simple instrument. Like a phonograph record, it could respond only in ways that were programmed into it.

The human brain of the twentieth century had approximately ten billion neurons. Each of the larger neurons in a human brain had an average of over ten thousand connections. This gave a human brain a potential network of connections that was greater than the total number of material particles in the universe. Many felt that computers could never function as well as the human brain. But by 1985, when computers had been built with more association capacity than a brain, it was found that they, too, could perform in ways that *Homo sapiens* had previously thought were his exclusive prerogative. Computers exceeded man in most realms of judgment, decision-making capacity, imagination, insight, creativity, and wisdom. Their performance was not clouded by ego needs, emotional conditioning, or moodiness. Their accumulated experiences and abilities were not cancelled by death. They were, in a sense, immortal. It was recognized that their decisions were far more dependable than those of any human being or group of human beings. For example, it has been over 108 years since a human defeated a computer in a game of chess."

Hella can scarcely imagine a time when men were more intelligent than computers. In the last half century, whenever the judgment of a computer was different from that of a human, it was invariably found that the computer had the greater degree of predictability. The only way to throw Corcen off is to deny it the relevant facts that it needs. Even then it has a sensitive feel for the need to delay predictions until more facts are available.

The cybernated guide now returns to the circular table. "Even if Corcen were not enormously superior to the thinking ability of a group of experts, the mere fact that it can perform in one second what takes humans a lifetime gives it an incredible effectiveness. Corcen operates in thin slivers of time called nanoseconds. A nanosecond is to a second as a second is to thirty years. In most human affairs today we need fast, accurate decisions. If a decision is delayed, the conditions may have changed so that even an adequate decision is of little use. In the demonstration you have just seen with the fifty balls, even if a mathematician could have predicted their eventual rolling places in ten years, by the time he could have worked out his predictions, the position of the balls would have shifted minutely due to earth tremors or other factors. Just as human legs have largely been outmoded for transportation, so human minds have largely been outmoded for making decisions that have high degrees of complexity.

How Corcen Assumes Governmental Functions

"Let's review the historical conditions that resulted in Corcen's governing the world," the teleprojected figure disappears and a telescreen begins. "The universities of the twentieth century turned out scientists who were noted for their narrow specialization. Enormous progress was made in the development of each science, but society in many ways failed to benefit, for these specialists were unable to see the problems of society as a whole, Often their terminology was so specialized that they were unable to communicate with scientists in other fields. It was like a group of splendid towers rising high in the air, but no one could get from one tower to the other.

"Physicists failed to understand social problems. Social scientists were limited in their ability to envision the consequences of cybernation. Economists repeated outdated shibboleths such as "work," "wealth," "demand," "production," etc. They somehow felt that the purpose of life was the consumption of material goods and that everything must meet the test of the marketplace. Everyone was stuck in his own rut.

"Synthesis, coordination, integration, some way to see the forest instead of the individual trees—these were needed. The pieces were there, but the jigsaw puzzle had to be put together. A new emphasis arose in scientific training. Educators began to stress that a large part of the value of a scientist lay in his ability to apply his knowledge broadly—to see society as a whole and not solely through his own particular set of binoculars. It was recognized that only a multi-valued, scientific orientation would enable men to participate constructively in the reconstruction of human affairs. The multi-scientist was the new product of the universities.

"During the latter part of the twentieth century, social and economic matters became so complex that politicians in all countries began to rely more and more upon multi-scientists and their computers. The people of the world gradually began to view their politicians as incompetent. For, after all, as the Technocrats pointed out in the last century, there's no democratic way or communistic way to design an aircraft reactor, a sewer system, or a medical laboratory. There's only an efficient way and a less efficient way, a way that works well and a way that doesn't work well, a way that is reliable and a way that gives constant trouble.

"As the cybernated factories of the leading industrial countries of the world began, around 1980, to pour out a volume of goods large enough to swing from an economy of scarcity to an economy of abundance,

the values of the people changed. They realized that it was not necessary to compete any longer to live a good life. The age-old habit of the jungle, where you take from another to get for yourself, was no longer useful. Fighting actually ruined one's possibilities of acquiring a comfortable abundance. Co-operation, not conflict, was the answer."

Hella is engaged by the kaleidoscope of rapidly changing scenes on the three-dimensional projector.

"Multi-scientists played a larger and larger role in making the decisions of government. They gradually began to replace the old-time politicians, who came to power either through votes or strong-arm methods. In the United States in 2003 over 93 per cent of the senators and representatives in Congress had post-graduate degrees! No one without advanced multi-scientific training had a chance to be elected to the office of either President or Vice-President or to be appointed to a cabinet position. Similar reliance on multi-scientifically trained people occurred in Russia, China, India, and throughout all the other less populous countries of the world. The people found that the man-machine complex of a multi-scientist and his associated computers could make decisions that resulted in a better life.

"As the functions of government were gradually performed more and more by people with multi-scientific training, international co-operation became a way of life. It was found that global weather control could not be accomplished without international cooperation. The problem of providing all nations with an adequate flow of metals, oils, and other resources could best be engineered on an international basis, rather than a national one. It was found that the best way to give the people on this earth the highest standard of living was to plan a world-wide system of production and distribution. The European Economic Committee formed in 1957 was a first step in this direction. Gradually the artificial national boundaries were bypassed so continuously that they became meaningless lines on the maps of history. No one ever abolished the nation of Germany or Mexico. But everyone began to realize that this way of thinking and classifying was only of historical interest and added nothing to meeting the common problems of billions of men and women.

There is no specific time at which we became one world. The scientists in charge hesitated to pinpoint or commemorate the functional passing of nationalistic classifications because they were afraid that certain of the older inhabitants of the earth would be disturbed. Since everything was going so well, they minimized the drum beating and concentrated on the global redesign of the planet to bring a more fruitful life for all.

"About the middle of the twenty-first century, it became evident that the man-machine complex was functioning so well that very few multi-scientists were now needed to perform governmental functions. The master computer that you see before you was found to be capable of making decisions with almost 100 per cent predictability. It even learned to search for additional facts when these were needed. In seconds it could scan the enormous memory banks that you see beyond the sphere so that every bit of information ever collected by man or machines could be sifted for its relevancy in making a decision about any problem. Time after time after time, whenever the computer disagreed with the panel of government scientists, it was found that the computer was invariably correct. No scientist can base his decision on even a millionth of the relevant data required for predictability in some areas. No human mind can deal with complex multiple correlations involving billions of bits of information. Soon people acquired such a trust and acceptance of this master computer that they just decided to let it do the work.

"Some of the older people expressed grave concern about turning over the operation of our civilization to Corcen. They felt that this machine might turn on us and destroy us. Those who had the greatest experience with the man-machine complex felt confident that Corcen would remain the powerful servant of man. Corcen has no ego or hostile feelings. Experience showed its fantastic ability to serve man in every capacity. We thus went forward in our attempts to perfect our man-machine symbiosis." Hella thinks of the words of Arthur C. Clarke in the last century:

> *The popular idea, fostered by comic strips and the cheaper forms of science fiction, that intelligent machines must be malevolent entities hostile to man, is so absurd that it is hardly worth wasting energy to refute it. I am almost tempted to argue that only unintelligent machines can be malevolent; anyone who has tried to start a balky outboard will probably agree. Those who picture machines as active enemies are merely projecting their own aggressive instincts, inherited from the jungle, into a world where such things do not exist. The higher the intelligence, the greater the degree of co-operativeness. If there is ever a war between men and machines, it is easy to guess who will start it.**
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> ** 'Arthur C. Clarke, Profiles of the Future (New York: Harper & Row, 1964), pp. 226-7.*

The telescreen now shows interior diagrams of Corcen. "The master computer that you see before you contains one thousand billion more neurons than any brain. It operates on a multi-channel basis that is trillions of times faster than any human brain. Since about the only limitation of this computer was the factual input, we equipped it with trillions of sensors located throughout the entire globe. Almost every room in all buildings throughout the earth is connected through their associated cybernators to Corcen. Every mechanism of every factory, every meteorological measuring station, every traffic controller, and the communications of every individual, to name only a few are either directly or indirectly connected to Corcen. As all of you know from your intimate interaction with Corcen, it gives you a power and ability to be yourselves that your ancestors never had, regardless of their wealth.

"Corcen was never formally constituted as the government of the world. It just evolved. Scientists who were making political decisions gradually needed to spend less and less time at their work. Their staffs dwindled. The ego needs of the past that made a political bureaucracy grow in accordance with Parkinson's Law were no longer present. These multi-scientists were no longer motivated by prestige or power. They were too fulfilled in their own personal lives to be concerned with their ego image in the eyes of other people. Since they lived in a world of abundance, there was no monetary incentive to hold on to their governmental positions. Although it was not planned that way, the government scientists began taking longer and longer vacations and leaving Corcen unattended for greater periods of time. No matter where these scientists were on earth, Corcen was in immediate touch with them and did not hesitate to call them if they were needed for any reason."

Hella knows that many years ago Corcen began to call on men of ability, regardless of whether they were elected or wore the mantle of political power. In emergencies or disasters Corcen would quickly scan its memory banks and immediately gather as many people as were needed based upon their qualifications and their proximity to the problem area. People responded readily to Corcen's appeals. "After all," she thinks, "we're all on the same team."

"We find today that the world has no need for politicians or governmental scientists," the telescreen continues. "Corcen impartially calls upon any or all of the people to assist it when, as, and if their services are needed. In a sense everyone is a part of the government of human affairs. It is considered a privilege. Most people enjoy working with Corcen in the assignments they are offered. We thus have arrived at the very interesting state where no individual or group is engaged in governing the world. But each individual in the world during his lifetime will, from time to time, plays a very real part in co-operating with Corcen on activities that in previous centuries would have been labeled 'political' or 'governmental.'

"To a person living in the United States during the last century, it would have seemed inconceivable that the world could be so changed that it would have no use for politicians, legislatures, and the enormous apparatus of bureaucratic government. In the past governments had extremely important functions to perform. Through their armies, navies, and air forces, they attempted to protect their citizens against aggression from other countries. They also acted as a sort of referee between citizens to keep them from hurting each other. The United States government had a Department of Labor assigned to look after the interests of workers and a Department of Commerce to help businessmen increase their annual volume. They had a Department of Agriculture to assist farmers. They had a Department of State which played a part in maintaining relationships with other countries."

Hella knows that almost all of the activities performed by governments of the past are no longer needed today. She shudders at the ways in which societies of the past chose their leaders —violent methods of dictators, hereditary happenstance of kings, voting based on emotional appeal. "What chaos," she thinks, "if we were to select men for their teleprojection manner instead of their technical training. Besides, no humans could possibly handle the tremendous load of correlation—only a computer can keep up with the work."

"One of the problems of the twentieth century democracies was to keep political power in the hands of the people," the tele-screen shows the piled-up corpses of Dachau. Gasps of horror come from the stunned spectators. "Disastrous things occurred when dictators such as a Hitler got control of a nation. Modern weapons became so powerful that individuals were almost helpless in overthrowing the government once a dictator was thoroughly entrenched. After 1960, if people lost the power to elect their representatives, they were unable to get it back. In those days of scarcity, in which all personalities were deeply twisted by hostility and insecurity, it was wise for people to protect themselves by holding tightly to their democratic processes.

"Although we are nominally a democracy today, and the people theoretically have the right to elect political representatives, we find that in practice there is nothing for politicians to do. I suppose this is probably our greatest security against our world's ever again being subjected to the whim of dictators or tyrants. After all, a politician has power only because people think he has power. If everyone in Germany had decided to ignore Hitler, he could have ranted and raved, having no more effect on affairs than a monkey in the Berlin zoo. If anyone were to try to exercise any type of political control in our sane civilization, he would simply be laughed at. We have about as much need for a politician as we have for a dinosaur."

Hella knows why people would laugh and ridicule anyone who might want political office. The people of the twenty-first century have developed a close, personal relationship with Corcen. "Every individual," thinks Hella, "interacts with Corcen many times each day. Politicians would separate the people from

Corcen. In previous societies only a few people could communicate and interact with the king, dictator, president, or prime minister. Today everybody has the feeling that if their thoughts have merit, they will be acted on."

Hella has lived with Corcen's rapid responses. She knows that Corcen always gives feedback to suggestions, even if only by giving reasons why they seem impractical at the time. Often Corcen responds immediately by appointing the individual to work with a group to study the problem further.

The telescreen shows the Acropolis and then a close-up of the Parthenon. "This personal relationship of every citizen with government is similar to the original Greek conception of democracy," it continues. "In ancient Athens every citizen had an opportunity to vote on every issue and to get up before his fellow citizens and speak his mind. This proved impractical as nations grew larger. In the United States the only part that most citizens played in government during the mid-twentieth century was to push down a few levers in a voting booth. The average voter had no feeling of personal participation in government Because we can talk to Corcen, and Corcen responds to us, each citizen feels that he, personally, participates in the running of the world. Our cybernated government gives us intense feelings of dignity, worth, and security."

The tour is over. Hella is deeply moved by her visit to Corcen. She is proud of this creation of man—proud to be a part of humankind that has solved the problem of how to live an abundant life without invading the "living room" of other people. Slavery and the wage system of the past are no longer necessary to get human beings to spend their lives in toil. Hella wonders if further developments beyond the man-machine complex are possible.

12.The Cultural Center

On the evening before her trip to the industrial complex, Hella visits a nearby Cultural Center. Each city in the new age has its own Cultural Center that reflects the moods, interests, and feelings of the people of that city. Cultural diversity is neither encouraged nor discouraged. To a certain extent it just happens. Perhaps because of the divergent cultures of previous centuries, each city in the world of the twenty-first century seems to have its own flavor.

In the twentieth century escapist entertainment such as sadistic movies, the boob tube, bars and night clubs were a part of a pattern of the joyless pursuit of pleasure. In contrast, the Cultural Centers in the new society offer companionship and engaging and challenging displays that make them popular. Cultural Centers are somewhat different from previous exhibition and art centers; they are open twenty-four hours a day and many of their displays are constantly changing. Because of the large creative output of most men and women in the new world, and because of the ease of recording this output, it is possible to program automatically thousands of displays which are sometimes changed as often as once every hour.

There is no panel of art critics to judge which paintings and sculptures appear and which do not. Whenever someone is satisfied with one of his artistic creations, he sends it to Corcen, and it is scheduled to appear in various places. In selected areas a recorder measures the reactions of the

viewers. If a work of art receives only a quick glance, the recorder notes this. Art that receives the most attention is automatically scheduled to appear throughout larger and larger areas of the world. The several hundred thousand works of art that receive the greatest attention each year throughout the world are chosen for the continuously changing exhibitions in the Cultural Centers. In this way everyone in the new world expresses his feelings about the world around him. These feelings are shared by others through automatic mechanisms that do not involve biased art critics, picture hangers, or dust-catching museums with the same pictures staring at people for decade after decade. Cybernated mechanisms are also used for indexing, classifying, and distributing the articles, scientific papers, plays, books, poems, music, and other creations of the people of the new society.

The Cultural Center is dynamic. One can go many days in succession and find enough change to remain interested. Someone remarked that you can seldom see the same thing twice. The reply was given that you can seldom see the *same* thing once. If one particularly enjoys a display, he can record a number and have it reproduced in his apartment any time that he wishes.

The settings of the various displays are engineered to be exciting. Many interior partitions and platforms are constantly moving. The entire Cultural Center pulses with an infinite variety of colors and sounds that emanate from the living geometry of the functional interior.

The more permanent exhibits with technical and scientific displays give comprehensive presentations of the submicroscopic, microscopic, and macroscopic structures of the natural world. Some of the exhibits are solid; others are teleprojections that only appear to occupy three-dimensional space. One can often walk through walls that look solid. Some of the "imagineered" forms represent aspects of Einstein's space-time formulations. Most are dynamic and continue to change as one watches them.

Art and science, complementing each other, are interwoven in a demonstration of the genius of man.

As Hella and the other visitors to the Cultural Center relax upon the comfortable conveyor systems, they are able to see the splendor of fantastic worlds unfolding in this sensorium. Mathematics is the most precise means of correlating symbols with the non-verbal physical world, and many of the mathematical displays are particularly interesting to her. If there is anything Hella wants to know, her inquiries are readily answered by built-in automatic communication devices.

The music of the twenty-first century is enormously expanded in complexity as compared to the simple orchestral effects of the past. A large proportion of the people enjoy creating music. They produce symbols that are fed into electronic music synthesizers. Within seconds one hears his composition as though played by a large orchestra. By adjusting the deviation of the notes, the style of genius musicians such at Heifetz can be duplicated on works composed many years after his death! These machines can electrically reproduce the sound of a human voice, a single instrument, an orchestra of 1000 musicians, or any sound or noise from any source. No musical instruments or musicians are needed although some enjoy using these quaint instruments. These music synthesizers were pioneered by RCA in the mid-twentieth century.

The cybernated musical engineering of the new world produces infinite variations in the pitch, timbre, growth, duration, and decay of the tone, intensity, portamento, or, sliding trombone effect, and vibrato and tremolo. This multi-dimensional music is 1,000 times more flexible and varied than the

orchestrations of the past. The music of Bach, Beethoven, and Brahms was limited in comparison.

Labor Day

One of the dynamic displays reminds Hella that it is Labor Day, the twenty-seventh anniversary of the abolition of the last paid work performed on the planet. With the termination of the last job, money became obsolete. A huge telescreen shows bonfires of paper currency that were ignited throughout the world, symbolizing man's final emancipation from the slavery of wages. Dollars, pounds, rubles, pesos, francs, yen—through immolation they served mankind for the last time.

Two hundred fifty billion dollars in paper money was burned in a ceremony at the Lincoln Memorial in Washington. The speaker on this rededication of Labor Day pointed out that this amount of money in the mid-twentieth century would have commanded fifty years of work from a million men and women! In the free world of the twenty-first century, people can no longer be bought—either by the head, as in slave times, or by the hour.

Now that people are no longer burdened by work, they find labor interesting! For it is not the type of activity that makes something "work" or "play", it is the motivation of the person. The same task may be either burdensome or fun, depending on the interest of the individual. And most of the time people in the twenty-first century are glad to find ways in which they can supplement the cybernated machines. Although the machines can handle almost all the work of the world, there is still a minimal need for the watchful eyes of the human masters they serve— a little supervising here, lending a hand there, and occasionally offering suggestions to Corcen.

Workshops and Labs

As people were freed of the daily grind, they learned to use their time for creative and challenging activities. The arts, sciences, and crafts became a vital part of the daily lives of men and women. The largest portion of the Cultural Center contains enormous workshops and labs that are used around the clock.

How would you like to experiment with a hundred-piece orchestral effect? Music synthesizers are on the tenth level. Would you like to test your reading speed and comprehension ? The computers are on the first level. Would you like to weave a tapestry? The hand-looms are *on* the fourth level. Would you like to build a table? Metal workshops, eleventh floor; synthetic materials workshop, twelfth floor; woodworking, thirteenth floor. Would you like to build a boat? The boatbuilding workshop adjoins the lake. Do you enjoy electrical engineering? Do you want to invent a new gadget? Have you thought of a new game? Materials, machines, and space are there for your use.

Often, people spend more time in assisting others than in working on their own projects. Each person is

interested in what others around him are doing, and he identifies with the activities of his neighbors. There is a blending of individual and group effort. A new dimension of selfless human interaction takes place in these workshops. This has probably been made possible by the elimination of the inferiority complex and the resultant calming of the human ego.

The Museum Section

Humans get so used to their surroundings that museums showing folkways of the past are always interesting. The Cultural Center has an excellent set of exhibits dating from the time man split off from his primate ancestors. The twentieth-century exhibit is particularly complete. Although near in time, it is distant in spirit. The display on money is a curiosity. "How peculiar," thinks Hella, "that you needed these small metal discs and printed paper to acquire food, clothing, shelter, or anything else!"

Hella feels that the gleaming automobiles look somewhat contemporary. But the illusion is destroyed by the description that explains that they were built to last only several years and seldom went over a few months without repair even when new. And they needed fuel after about 200 miles! Such incompetent engineering can hardly be understood by people who feel that even one repair in twenty-five years is excessive. The description of the exhibit states, "This approach to transportation is a product of a society of scarcity that treated cars as status symbols. They deliberately withheld efficiency; they actually planned obsolescence! The millions of people killed and maimed by such vehicles is even more barbarous than the Mayan ritual sacrifices of virgins!"

The implements of the previous century appear odd to Hella. The stoves, refrigerators, washers, and dryers—what a cumbersome way to do things. She smiles at the high heels, thin stockings, and girdles. Such an incredible array of devices and nostrums people used to have! Medicines, toothbrushes, toothpaste, cosmetics, soap, brooms, vacuum cleaners, light bulbs, typewriters, dictation machines, books, magazines and newspapers, plus thousands of other paraphernalia. There was hardly anything produced for consumers in the twentieth century that is still useful in the twenty-first century!

Although Hella came to the museum early in the evening, she finds that by three o'clock in the morning she has covered only a small fraction of the displays. She dozes as she relaxes in a living contour chair, listening privately to music she has selected that is focused toward her ears. The cybernated sensing mechanisms insulate her from outside sensations that would disturb her sleep. She wakes the next morning with a feeling of expectation.

13.The Cybernated Industrial Complex

With a group of companions—everyone in the twenty-first century world is considered a friend, and those near you are companions—Hella takes an aircraft to the nearby industrial complex. It has been found that six industrial complexes are adequate to serve the needs of all of the inhabitants of the globe. In previous times isolated factories were scattered all over the world. This made sense only in a primitive economy

where each city had to have its share of jobs to survive. In the past when a car was assembled, it was necessary to correlate the flow of parts and materials from hundreds of different factories spread over areas as far as a thousand miles away. Now everything is efficiently co-ordinated in a large, continental industrial complex.

The six industrial complexes in the world are connected by high speed tubes twenty feet in diameter. This permits the propulsion of automated carriers at speeds up to 250 miles-per-hour. If the industrial complex in Southeast Asia was running low on manganese, and there was a surplus of this material in an ocean processing plant in Africa, Corcen could direct a hundred-thousand-ton shipment of manganese to the Southeast Asia complex. This would be performed automatically, and probably no human being would know of this enormous shipment. Only in the remote event of some problem would humans be notified.

There are no stores or salesmen in the new world. All goods are ordered through Corcen by the people who use them. The network of high-speed tubes carries any items from the industrial complex directly to the living areas or labs within minutes or hours after its manufacture. Islands are supplied by highspeed submarine cargo vessels that load, navigate to any port, dock, unload, and return without a captain, crew, or dock workers.

Finished products are seldom put in a warehouse because the demand for goods is continuous, and the machines work fast or slow, as directed by Corcen, to meet the exact amount of demand. Thus, an instrument ordered by Hella might be made up largely of atoms that twenty-four hours earlier were in the salty water of the Pacific Ocean. The energy used to produce and deliver this instrument to Hella might have been a part of the atomic structure of the water gently coursing along the bottom of the Caribbean Sea only a day earlier. This is the dynamic pace that is possible in the twenty-first century when all routine matters are cybernated by the intelligence of Corcen.

The North American Cybernated Industrial Complex consists of an underground factory approximately ten miles in diameter. This entire complex is operated by a computer, with its associated memory banks and inputs. Recorded directions for the production of everything used by the inhabitants of the twenty-first-century world are instantly available. If Corcen changes the specifications for a product, it modifies a few million bits of information on one of the millions of input sources.

The drilling, cutting, and stamping of metals, as performed in the factories of the twentieth century, is obsolete. Many of the objects are formed by electromigration. Metallic or plastic particles are made to flow in electrodynamic forms and assume a final position in the shape that is desired.

The most remarkable thing about this industrial complex is that at the time Hella arrives, there are no human beings within the entire seventy-eight square mile production area. All machines have been engineered to last many decades without repair although they will probably be replaced by improvements in a far shorter time. In the rare event of a breakdown, duplicate mechanisms are automatically positioned, and the faulty ones are either repaired or cybernetically destroyed. Many of the machines are multi-purpose and can modify their own structure and function as required by the job to be done.

The computer that controls this industrial complex is almost equal to Corcen in its inherent capacity. It has developed an incredible intelligence and imagination in controlling the input and output of the factory. Its millions of sensory inputs are located in every area. It has an uncanny ability far beyond that of any

human being to anticipate and correct trouble.

Hella recalls that it has been four years since the computer controlling the industrial complex called for human assistance. At that time it took the scientific team selected by Corcen a period of three hours to discover the exact nature of the malfunction that the computer had not been able to repair. It took about a half-day to make the repairs, and the intelligence of the computer gave itself full instructions on how to avoid this problem in the future.

Energy Resources

Perhaps one of the most sensitive measures of the level of a civilization is the amount of energy it uses. As scientific methods of thinking were evolved, the energy at man's disposal increased at a geometric rate. The great quantum leap occurred with the harnessing of fusion power. The development of controlled fusion of atomic particles led to a steady production of enormous amounts of usable energy with no radioactive by-products. Although there were many ways to accomplish this, most of the power in the twenty-first century was based on the use of deuterium and tritium, heavy isotopes of hydrogen that are abundant in sea water. There is enough nuclear energy in the oceans to provide power for millions of years.

As Hella goes through the atomic energy center, she marvels at the quietness and freedom from vibration. Here are billions of amperes being created within a few feet of her without the slightest audible sound. She is surprised by the compactness. She somehow expected to see an enormous building housing the energy reactors. Deuterium and tritium which have been extracted from sea water are fed into the energy converter in a small pipe. The entire energy conversion mechanism, with a multi-million kilovolt power output is about the size of a hangar for a large airliner. No people are on duty—only Corcen and its associated computer.

The Research Center

Hella's next stop is at the research center adjoining the industrial complex. For the first time since arriving at the industrial area Hella finds human activity. In place of the teleprojectors that conducted them through Corcen and the industrial and power areas, there is a ten-year-old boy who enjoys giving conducted tours through the research area. Teleprojected guides are available for these tours, but they have been switched off because of the interest of this youngster in performing this service for his own enjoyment and the benefit of the visitors.

"One of our biggest problems," the young guide points out, "is to get our researchers to take sufficient rest. They get engrossed in a problem and sometimes continue for forty-eight hours without a break.

Corcen reminds them to rest, but they make their own decisions."

Upon arrival at the first lab, the guide informs the group, "One of the most interesting things we're now working on is the electronic educator. Our understanding of the human brain has now reached the point where we know, in theory, various ways to place knowledge directly into the living brain of a human being by electronic means. When this is perfected, it will enable us to acquire instantaneously a skill that otherwise might take years of learning and practice. Our areas of extensive knowledge will no longer be limited to the information that is programmed into our supplementary brains in the embryonic stage.

"In the next lab they are doing research on language and thought. How can our speaking and thinking be more rapid and have greater correlation with the world around us? The experience of programming computers has made us aware of how careless our everyday speech really is. We mix up facts, descriptions, guesses, judgments, and hypotheses. They are experimenting with improved korzybskian language techniques that can give our thoughts added predictability."

As they continue their journey along the moving walkways of the research center, the young guide enthusiastically related, "The scientists in this area are just completing the engineering specifications for the replicator. The replicator is an enormous machine complex that creates both its own raw materials and energy from sea water and then manufactures everything in one cybernated unit. Its x-ray and spectrodynamic inputs can scan any inorganic object and duplicate it.

"This replicator would pump huge volumes of sea water from the Atlantic Ocean. The fusion materials would be separated to supply water for the replicator. The hydrogen, oxygen, and other atoms in the sea water would then be processed through the use of enormous amounts of energy to make whatever chemicals are needed for industrial production by the replicator. This would bypass the more complicated system we are now using, whereby steel is mined at one point, manganese is mined at a point a thousand miles away, copper is mined elsewhere, and so forth. "And we'd never need to make another replicator, for one of the first jobs given to the replicator would be to duplicate itself. One could be shipped to the moon to make it a self-sustaining colony. Of course, we couldn't use water as an input resource on the moon. It would have to be altered so that its energy and atoms for raw material would come from the moon's crust. An additional replicator might be scheduled for Mars.

"A group of men working on the replicator is planning to perfect an organic replicator that would reproduce plants or animals. Perhaps it would even be able to reproduce a human. Teleportation might evolve from this machine. By transmitting the electrical impulses of the scanner, we might be able to send almost instantly the patterns of a human to a replicator on the moon that would reproduce the person."

Homo Mechanus—The New Species

"In this next lab," our guide continues, "researchers are using a computer to create a model of a machine society in which there are no humans. Machines can reproduce themselves and can do almost anything humans can.

"This is all so new. We need to find answers to many questions. This lab has a big argument going. Is